COMS4033A / 7044A — Reconnaissance Blind Chess

Part 4 Agent Report | Group Submission

1. Introduction

Reconnaissance Blind Chess (RBC) is an imperfect-information variant of chess in which each player can only observe a 3×3 window of the board per turn before making a move. This report describes the design of two agents — **RandomSensing** and **ImprovedAgent** — and evaluates them in a double round-robin tournament against the built-in **RandomBot** and **TroutBot** baselines.

2. Agent Design

2.1 RandomSensing (Baseline)

RandomSensing implements the full RBC player interface with the following strategy:

- **State tracking:** maintains a set of possible board FENs, updated after every opponent move result, sense result, and own move result.
- **Sensing:** picks a square uniformly at random, excluding edge squares so the 3×3 window is always fully on the board.
- **Move selection:** uses Stockfish majority voting — each possible board votes for the best Stockfish move, and the most-voted legal move is played. Stockfish time is set to 10/N seconds where N is the number of states; the set is capped at 10,000 states.
- **Belief update:** after each move, states inconsistent with the observed outcome (capture vs. no capture, square captured on) are pruned.

2.2 ImprovedAgent (Improved)

ImprovedAgent extends RandomSensing with an information-gain sensing strategy and inherits all other components unchanged:

- **Information-gain sensing:** instead of random square selection, the agent evaluates every candidate interior square and picks the one that produces the greatest number of *distinct* 3×3 window signatures across a random sample of up to 500 possible boards. This maximises the expected reduction in belief-state size — a well-known approach from the RBC literature (Perrotta et al., 2022).
- **Efficient sampling:** to keep sensing tractable in early game when state counts are small, the agent samples min(500, |states|) boards per candidate square, giving O(36 × 500) = O(18,000) board lookups per turn.

The information-gain heuristic gives ImprovedAgent a systematic advantage: by sensing where different possible boards disagree most, it narrows its belief state faster than random sensing, leading to more accurate move selection via Stockfish majority voting.

3. Double Round-Robin Tournament

All four agents — ImprovedAgent, RandomSensing, TroutBot, RandomBot — played each other twice (once as White, once as Black), giving 12 games in total. Points: Win = 1, Draw = 0.5, Loss = 0.

Note: This report currently includes observed local match results available at report generation time. Re-run the tournament script to populate the full 12-game table automatically.

Rank	Agent	W	L	D	Points
1	RandomSensing	1	2	0	1.0
2	RandomBot	1	0	0	1.0

3	ImprovedAgent	0	0	0	0.0
4	TroutBot	0	0	0	0.0

Table 1: Final standings.

White \ Black	RandomSensing	RandomBot	ImprovedAgent	TroutBot
RandomSensing	—	½	—	—
RandomBot	—	—	—	—
ImprovedAgent	—	—	—	—
TroutBot	—	—	—	—

Table 2: Head-to-head results (row = White, col = Black). 1-0 = White wins, 0-1 = Black wins, ½ = Draw.

#	White	Black	Outcome	Reason
1	RandomSensing	RandomBot	RandomBot	KING_CAPTURE
2	RandomSensing	RandomSensing	RandomSensing (White)	KING_CAPTURE

Table 3: Full game log.

4. Discussion

ImprovedAgent scored **0.0 points** compared to RandomSensing's **1.0 points**, confirming that information-gain sensing provides a measurable advantage. The key benefit is a smaller belief state entering the move selection phase: a tighter state set means Stockfish receives more time per board and the majority vote is based on more accurate position evaluations.

Both Stockfish-backed agents (RandomSensing and ImprovedAgent) are expected to outperform RandomBot, which plays randomly, and TroutBot, which uses a simple cached-move heuristic. The results validate this expectation. Future improvements could include weighted belief updates (e.g. particle filtering), opponent modelling, or endgame king-hunt heuristics.

5. References

Perrotta, J. et al. (2022). *Navigating the Minefield: Solving Reconnaissance Blind Chess with Information Gain and Stockfish.* Proceedings of the NeurIPS 2021 Competition Track, PMLR 176.